LISA | TWO-BELT DRIVE R06 FIT SET

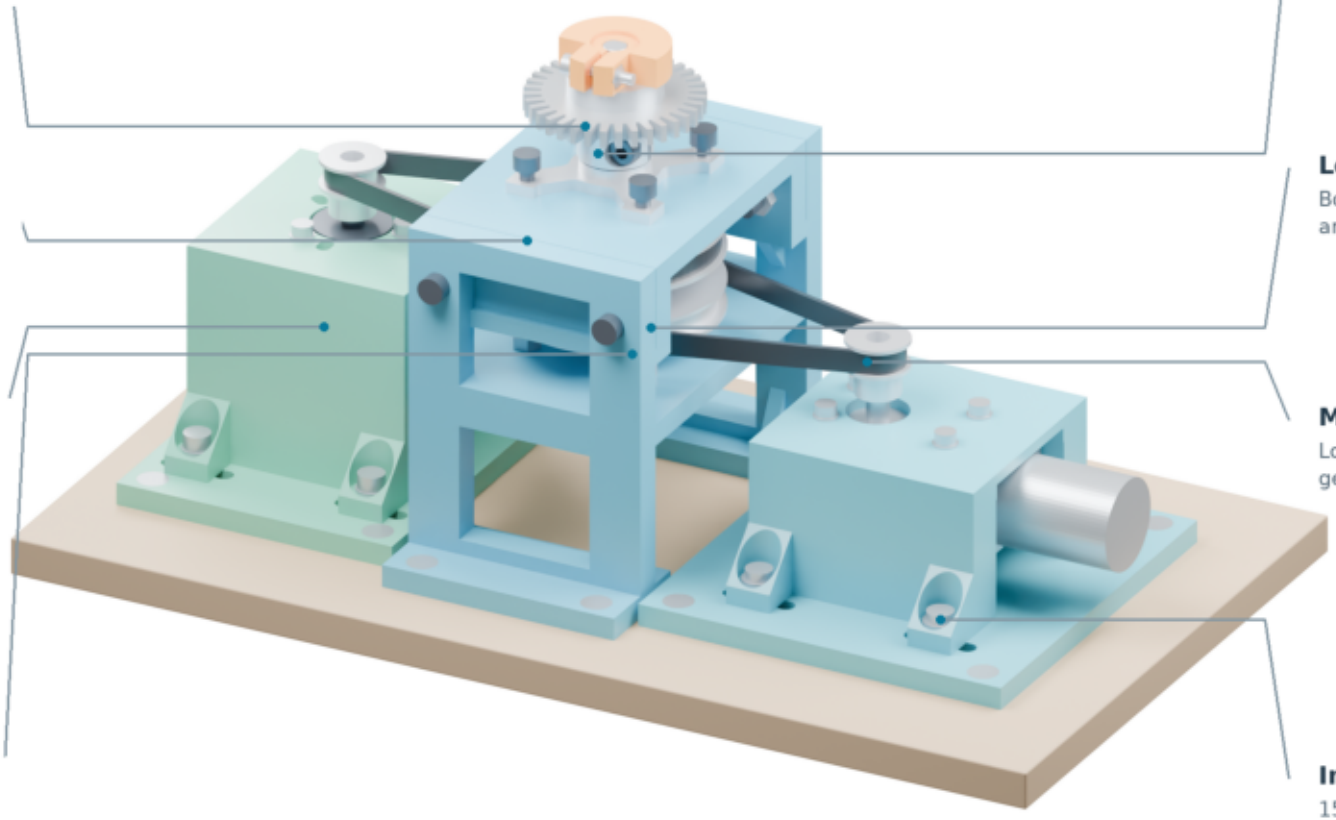
Motor → supported gear shaft → encoder. Two 250 mm x 6 mm GT2 belts; both devices adjustable.

Existing gear
Above the upper bearing;
printed keyed adapter

Upper 80T → 20T
4x encoder speed
relative to the gear

Encoder + 20T
Runs at motor RPM;
face raised by 16 mm

Integral shaft frame
Twelve local gussets;
removable upper bridge



Exposed upper collar
Above the bearing,
outside the plastic plate

Lower 80T
Both large pulleys
are between bearings

Motor + 20T
Lower belt reduces
gear speed to 1/4

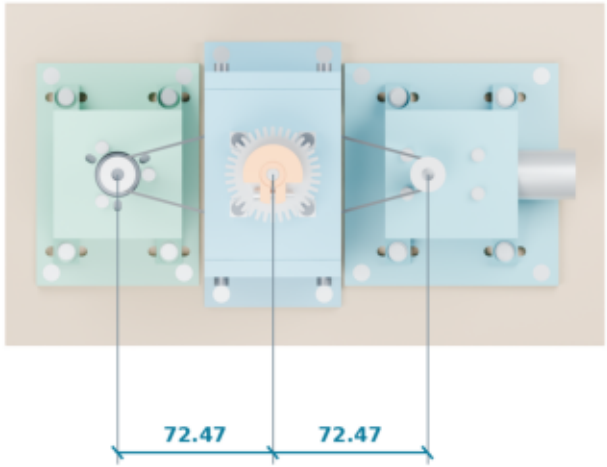
Independent slides
15 mm motor travel;
15 mm encoder travel

Motor 20T → lower 80T: gear at 1/4 motor RPM. Upper 80T → encoder 20T: encoder at motor RPM.
Printable fit set: check actual pulley heights, gear key fit, shaft grip and loaded stiffness before use.

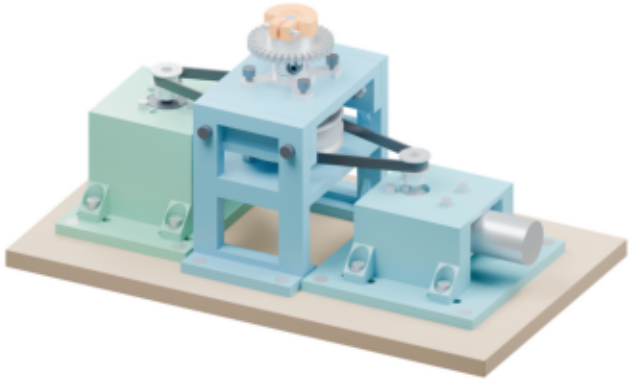
LISA | TWO-BELT LAYOUT — R06 FIT SET

Nominal dimensions in mm. Four views of the same FreeCAD assembly.

TOP | 250 mm belts; separate tension adjustments



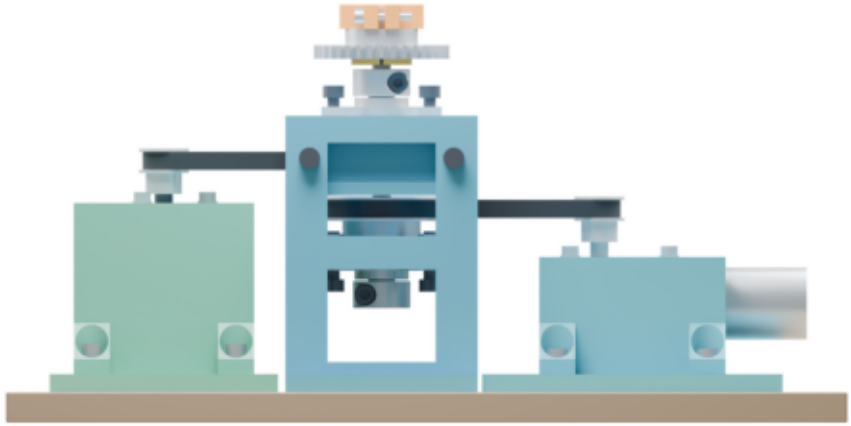
ISOMETRIC | exposed motor, separate gear shaft



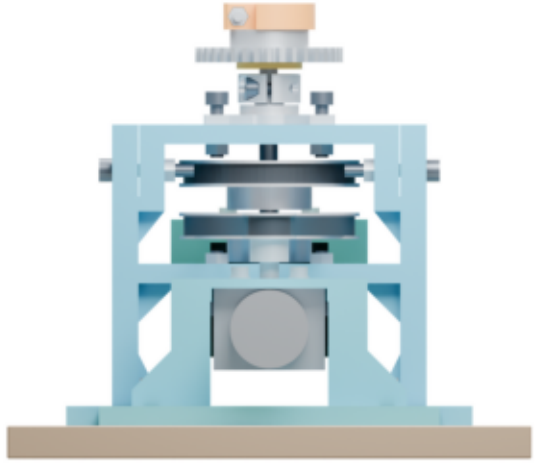
Gear: 1/4 motor RPM | Encoder: motor RPM

Two 80T pulleys on one 100 mm steel shaft

FRONT | belt heights: 61 / 77 above plywood



SIDE | integral frame and removable upper bridge



Bearing centres: 40 / 95.5 above plywood (55.5 span). Steel shaft: 28–128 above plywood.

R06 printable fit set: pulley axial dimensions, gear key fit and loaded stiffness need physical checks.

LISA | BEARINGS AND COLLARS — R06 FIT SET

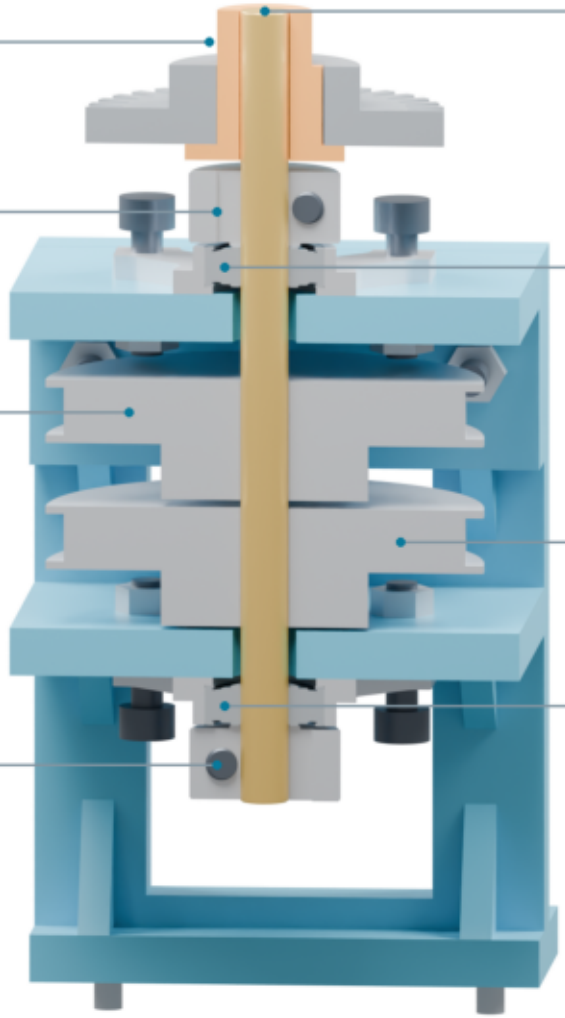
Actual CAD section: front halves removed. Gold shaft is kept whole. Both collars remain outside the plates.

Gear adapter clamp
M3 x 16 + captive nut;
removable sleeve clamp

Upper shaft collar
Accessible above
the upper bearing

Upper 80T pulley
77 mm belt centre
Encoder speed boost

Lower shaft collar
Front window gives
access to the pinch bolt



100 mm steel shaft
6 mm D profile
28–128 above plywood

Upper bearing
On the removable
upper bridge

Lower 80T pulley
61 mm belt centre
Motor speed reduction

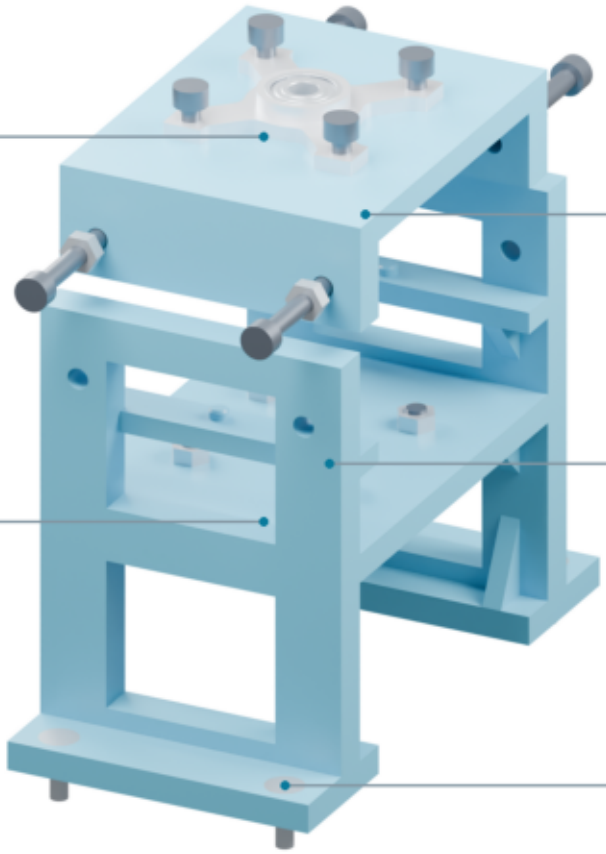
Lower bearing
Under the integral deck;
55.5 mm bearing span

Bearing housings bolt to the plates. Collars clamp to the shaft and lightly contact the bearing inner rings.
Bearing blocks and collars use manufacturer STEP geometry. Pulley and gear envelopes remain provisional.

LISA | BENCH ASSEMBLY — R06 FIT SET

Fit the lower bearing before board mounting. Load both pulleys and belts, then lower the upper bridge.

Upper bearing + bridge
Prefit four M4 x 16;
lower over bare shaft



Removable upper bridge
Four M4 x 25 side bolts;
seats and pins locate it

Lower bearing + deck
Slide bearing in sideways;
four M4 x 16 screws

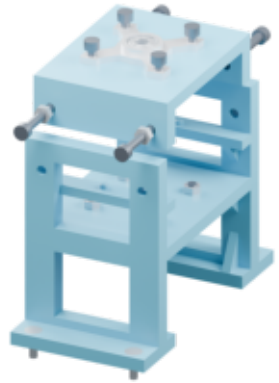
One rigid frame body
Twelve gussets replace
old screw-joint ledges

Integrated feet
Four #6 x 1/2 screws;
mount after bearing fit

Exploded view shows only the stationary frame and bearings. Shaft, belts and devices omitted for clarity.
Add collars and gear after the bridge is seated. Start all four side bolts before tightening evenly.

Assemble the bearing sandwich on the bench

Two printed frame pieces • Fit both closed belts before lowering the upper bearing bridge



Gusseted body stays fixed; upper bridge lowers along the shaft.

Frame hardware / components	Quantity
Frame body	1
Upper bridge	1
Bearing blocks	2
M4 x 16 bearing screws	8
M4 x 25 bridge screws	4
Plain M4 nuts	12
6 mm shaft	1
Shaft collars	2
80T pulleys	2
Closed GT2 belts	2

1. Remove every support under the lower deck and inside the round bores. Before board mounting, slide the lower bearing in through an open X side and bolt it underneath the integral deck with four M4 x 16 screws and four nuts. Driver access is from below.

2. On the bench, attach the upper bearing above its bridge with four M4 x 16 screws and four nuts. Leave the bearing fasteners just loose enough to align with the shaft; omit extra bearing washers.

3. With the bridge off, lower the bare shaft and both 80T pulleys into the lower bearing. Loop both closed belts around their pulleys now. Leave the upper collar, gear and adapter off so the upper bearing can pass over the bare shaft end.

4. Lower the bridge and upper bearing vertically over the shaft. Guide the front pin into the round socket and the rear pin into the slot. Seat both flanges on their shoulders. Do not use bolts to pull a misaligned bridge down.

5. Insert four M4 x 25 screws from the outside front and rear. Hold the four M4 nuts with a 7 mm spanner through the open X sides. Start all four screws, then tighten evenly; align and secure the bearing mounts while checking free shaft rotation.

6. Fit the lower collar from below and the upper collar from above; set light contact against the bearing inner rings without preload. Fit the gear adapter, then mount the frame to plywood. Align pulleys and tension both belts using the device slides.

No adhesive. Locator fit assumes 0.25 mm radial clearance and 0.25 mm end clearance. CAD tool checks use a 5 mm driver stem and a 3 mm thick, 14 mm OD wrench head; check your tools and printed fit before tightening.

Purchased parts: goBILDA first

R06 fit set • Individual quantities • Blue part numbers open the supplier pages

Component	Quantity	Specification	goBILDA part number	Each	Subtotal
Steel D shaft	1	6 mm diameter, 100 mm length; stainless steel	2101-0006-0100	$4.29	$4.29
Single-bearing block	2	6 mm bore; 32 x 32 mm mounting pattern; bearing included	1603-0032-0006	$6.99	$13.98
Clamping shaft collar	2	6 mm ID, 19 mm OD, 9 mm body length; M4 pinch bolt included	2910-0919-0006	$4.49	$8.98

Shaft, bearing blocks and collars: USD 27.25

Each bearing block includes its bearing. Each collar includes its M4 pinch bolt. The blocks and collars use manufacturer STEP; the steel shaft also uses the supplier CAD in this assembly. The quoted subtotal excludes other fasteners, belt, filament, shipping and tax. Prices checked 2026-10-04.

Component	Quantity	Source
Motor	1	Reuse existing component
Encoder	1	Reuse existing component
20T pulley	2	Reuse existing component
80T pulley	2	Reuse existing component
Timing belt	2	Both belts on hand (user report)
Driven gear	1	Reuse existing component
Plywood board	1	Reuse existing component

Required belts: closed-loop 250 mm pitch length, 6 mm width, 2 mm pitch, 125 teeth. User reports components are on hand; check actual specifications and fit.

Complete screw, nut and washer list

Quantities cover the whole r06 assembly. Use stock already available where it matches the listed specification.

Location / component	Quantity	Specification	goBILDA part number
Bearing mounting screws	8	M4 x 16 mm socket head	2800-0004-0016
Frame joining screws	4	M4 x 25 mm socket head	2800-0004-0025
Motor slide clamp screws	4	M4 x 12 mm socket / button head	2800-0004-0012
Encoder slide clamp screws	4	M4 x 12 mm socket / button head	2800-0004-0012
Plain M4 hex nuts	20	M4 x 0.7; 7 mm across flats	2811-0004-0007
Slide clamp washers	8	M4 flat washer; approximately 1 mm thick, OD at most 10 mm	2801-0004-0008
Motor mounting screws	4	M3 x 6 mm round / pan / button / socket head	Existing / generic
Encoder mounting screws	3	M3 x 8 mm round / pan / button / socket head	Existing / generic
Device mounting washers	7	M3 flat washer; approximately 0.6 mm thick	Existing / generic
Plywood wood screws	12	#6 x 1/2 inch (12.7 mm), flat head	Existing / generic
Gear adapter clamp screw	1	M3 x 16 mm	Existing / generic
Gear adapter clamp nut	1	M3 hex nut	Existing / generic

Both slides use M4 x 12 screws with 1 mm washers on 5 mm feet, above 6 mm bases. Captive nuts have full nominal 3.2 mm engagement and the screw tips reach the plywood surface. The selected goBILDA washer is confirmed 1 mm thick in supplier CAD.

Use plain M4 nuts at the bearing plates and omit extra washers in that stack. Longer upper bearing bolts can contact the upper pulley. The 20 M4 nuts cover 8 bearings, 4 bridge joints and 8 slide clamps. Twelve #6 x 1/2 inch wood screws secure the two bases and the frame feet.

Motor M3 x 6 and encoder M3 x 8 give about 2.4 mm engagement with 0.6 mm washers; verify actual thread depths. Encoder M3 x 10 plus a rigid 2 mm spacer is the alternative. Gear cap: one M3 x 16 screw and one M3 nut in the retained hex pocket.

Print settings and PLA estimate

Bambu Lab A1 mini • 0.4 mm stainless steel nozzle • All 8 STLs exported in printing orientation

PLA: 0.20 mm layer height; 6 walls (about 2.5–2.7 mm); 6 top and bottom layers (1.2 mm). Use 40% gyroid for structural parts and 100% infill for both gear-adapter pieces. An optional 3 mm brim fits within the A1 mini envelope.

Support the underside of the integral lower bearing deck and the eight flat holder washer lands. Add local supports in horizontal M4/M3 bores and at the clamp nut-pocket roof as needed. Remove all support before assembly. Part pages show exact orientations.

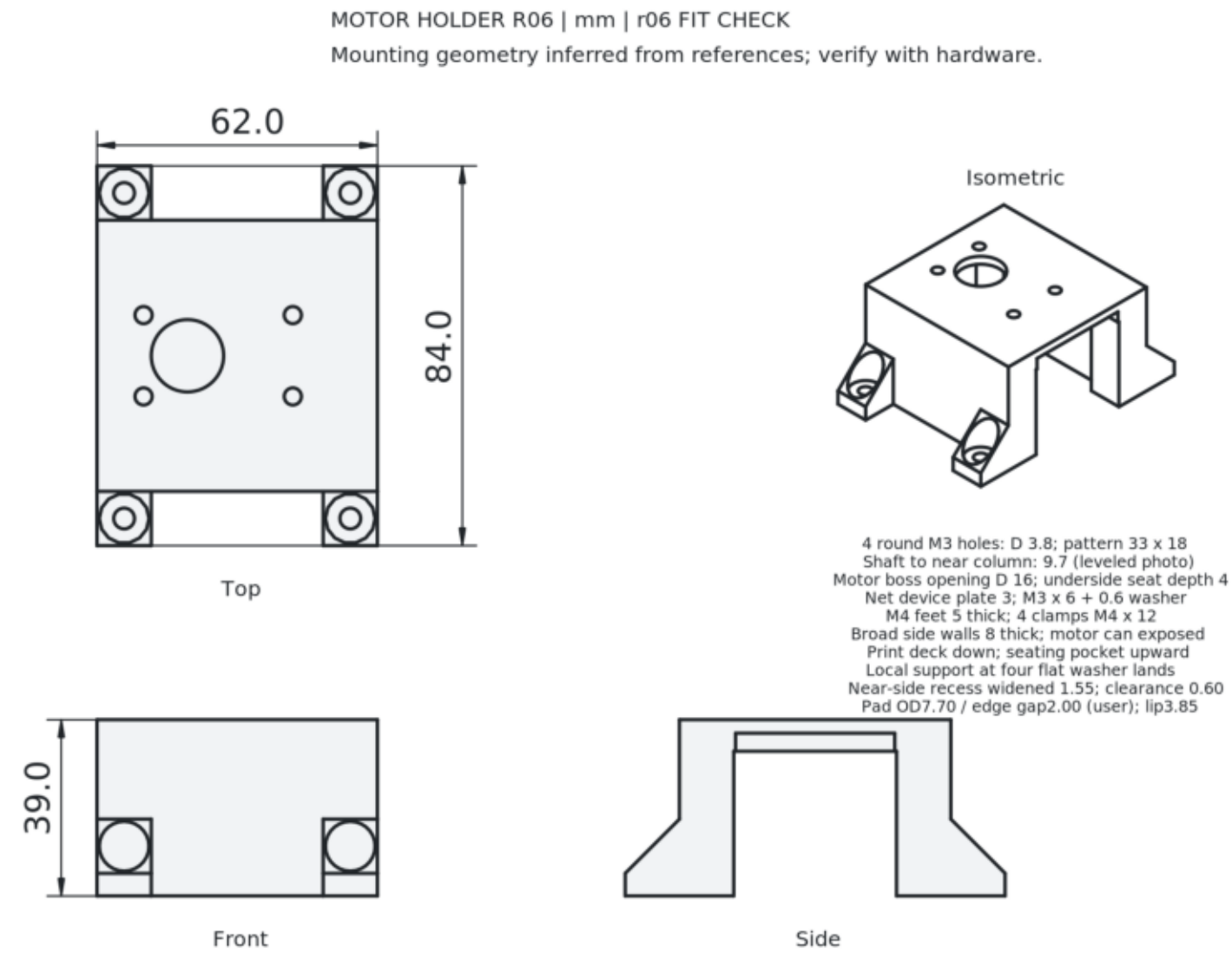
Printed component	Quantity	PLA	Filament cost	Infill
motor holder	1	57.9 g	$1.16	40%
motor slide base	1	52.3 g	$1.05	40%
encoder holder	1	78.5 g	$1.57	40%
encoder slide base	1	39.7 g	$0.79	40%
gear adapter sleeve	1	2.5 g	$0.05	100%
gear adapter clamp cap	1	4.0 g	$0.08	100%
shaft frame body	1	108.7 g	$2.17	40%
shaft upper bearing bridge	1	42.1 g	$0.84	40%

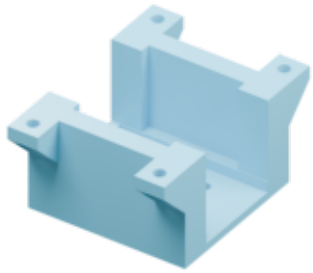
Complete print set: approximately 385.7 g PLA / USD 7.71

Geometric estimate, not a slicer result: PLA USD 20/kg, density 1.24 g/cm³, 5% waste. Shells estimated on a 0.6 mm voxel grid; allow approximately ±20%. Slicer-generated support, TPU, hardware, electricity and labor are excluded.

Motor Holder R06

Print one • Native FreeCAD TechDraw: three orthographic views + isometric • Drawing shows local assembly orientation





PRINT ORIENTATION

Top deck on bed; feet and underside housing seat upward

PLA: ~57.9 g

Filament: ~USD 1.16

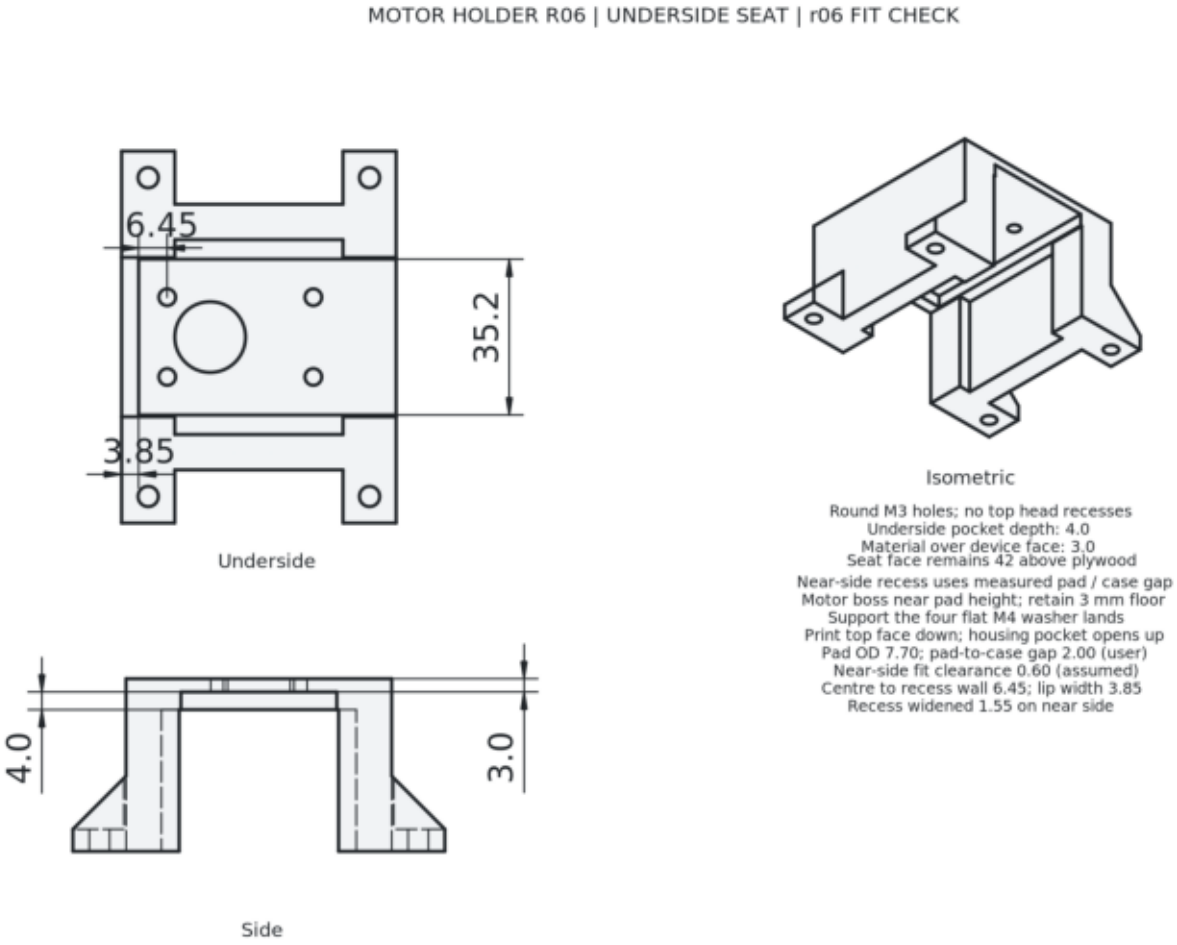
Infill: 40%

Local support at four M4 washer lands in the foot driver chambers

STL: motor holder/motor_holder_r06.stl

Motor gearbox seat: near-side relief

User dimensions locate the gearbox edge; 0.60 mm clearance is the selected printing allowance

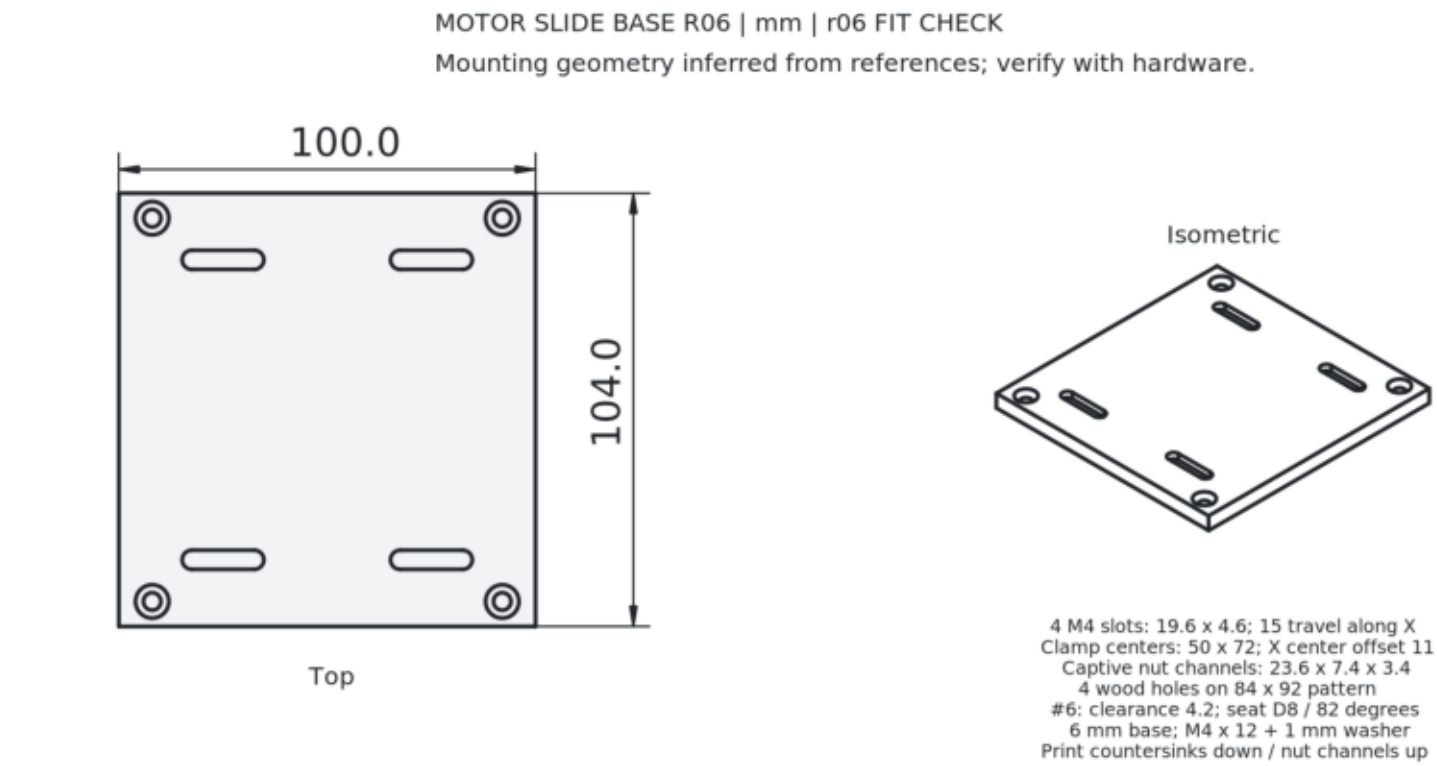


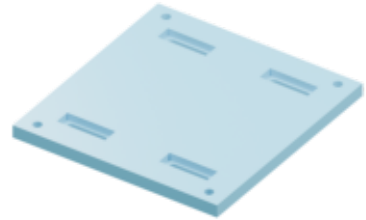
The raised circular mounting pad is 7.70 mm outside diameter, with 2.00 mm from its edge to the gearbox edge (user measurements). The gearbox edge is therefore 5.85 mm from the screw centre. The revised pocket wall is 6.45 mm from that centre, providing 0.60 mm nominal clearance.

The near-side recess is widened by 1.55 mm, leaving a 3.85 mm lip. Pocket depth remains 4 mm and the screw-bearing plate remains 3 mm. Dry-fit the revised holder against the actual gearbox before tightening its screws.

Motor Slide Base R06

Print one • Native FreeCAD TechDraw: three orthographic views + isometric • Drawing shows local assembly orientation





PRINT ORIENTATION

Countersinks toward bed; captive nut channels upward

PLA: ~52.3 g

Filament: ~USD 1.05

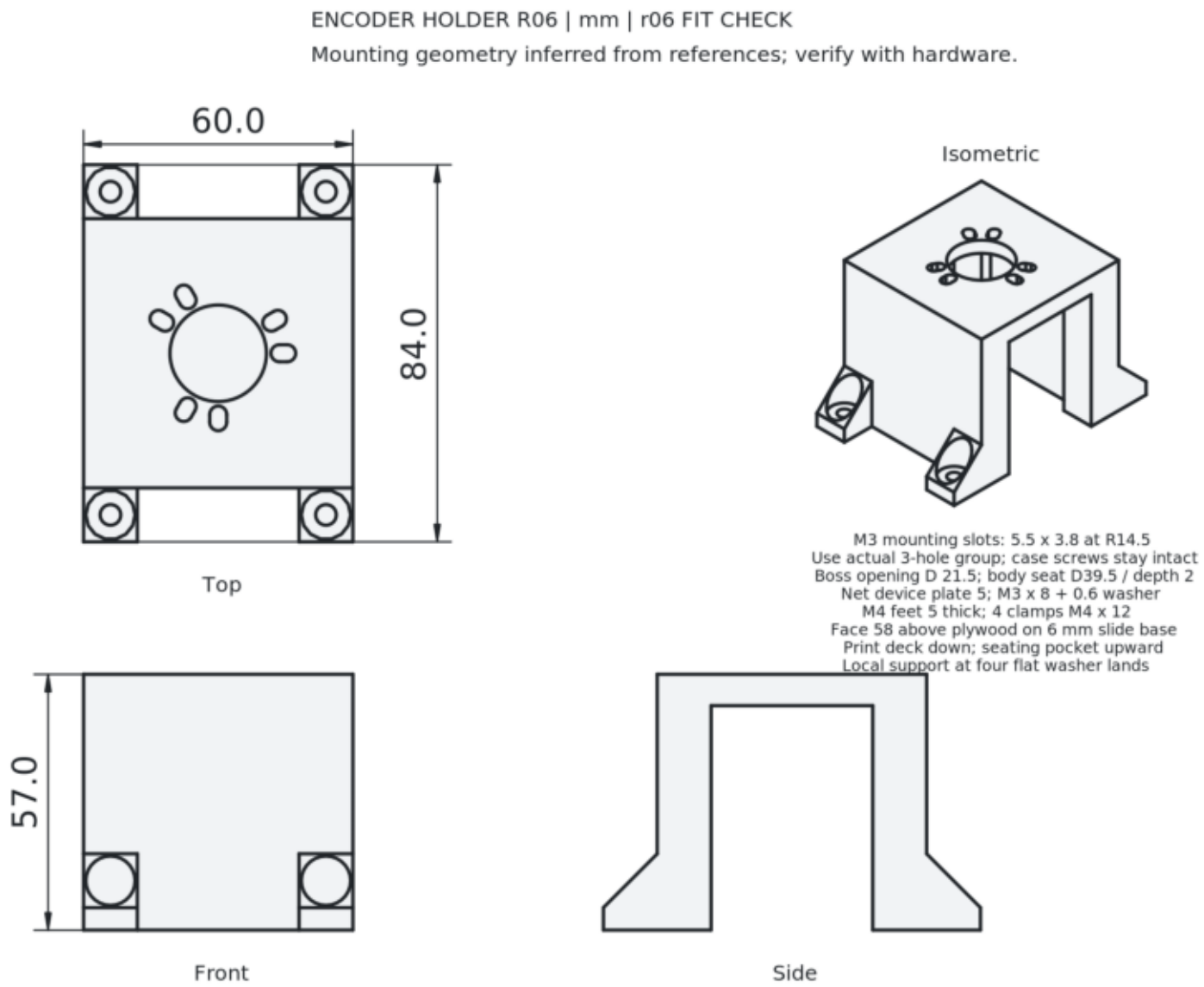
Infill: 40%

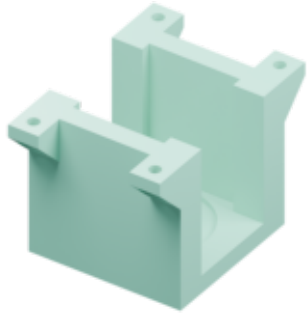
None intended; inspect slicer preview

STL: motor holder/motor_slide_base_r06.stl

Encoder Holder R06

Print one • Native FreeCAD TechDraw: three orthographic views + isometric • Drawing shows local assembly orientation





PRINT ORIENTATION

Top deck on bed; feet and underside housing seat upward

PLA: ~78.5 g

Filament: ~USD 1.57

Infill: 40%

Local support at four M4 washer lands in the foot driver chambers

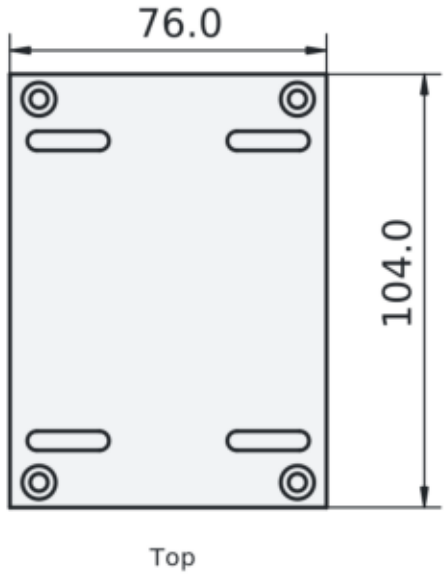
STL: encoder holder/encoder_holder_r06.stl

Encoder Slide Base R06

Print one • Native FreeCAD TechDraw: three orthographic views + isometric • Drawing shows local assembly orientation

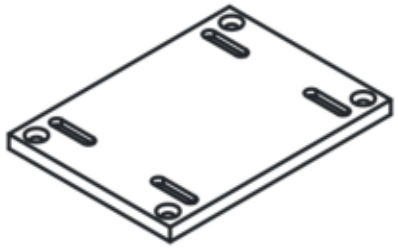
ENCODER SLIDE BASE R06 | mm | r06 FIT CHECK

Mounting geometry inferred from references; verify with hardware.



Top

Isometric



4 M4 slots: 19.6 x 4.6; 15 travel along X
Clamp centers: 48 x 72
Nut channels: 23.6 x 7.4 x 3.4
4 wood holes on 62 x 92 pattern
#6: clearance 4.2; seat D8 / 82 degrees
6 mm base; M4 x 12 + 1 mm washer
Fix base to wood before fitting encoder holder





Front



Side

PRINT ORIENTATION

Countersinks toward bed; captive nut channels upward

PLA: ~39.7 g

Filament: ~USD 0.79

Infill: 40%

None intended; inspect slicer preview

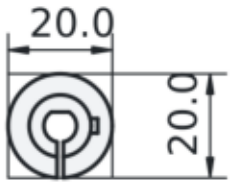
STL: encoder holder/encoder_slide_base_r06.stl

Gear Adapter Sleeve R06

Print one • Native FreeCAD TechDraw: three orthographic views + isometric • Drawing shows local assembly orientation

GEAR ADAPTER SLEEVE R06 | mm | r06 FIT CHECK

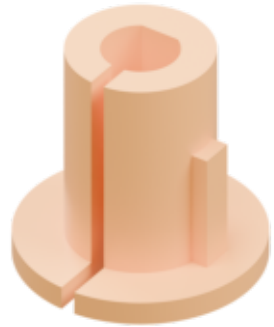
Mounting geometry inferred from references; verify with hardware.



Top

Isometric





PRINT ORIENTATION

Support flange on bed; keyed sleeve upward

PLA: ~2.5 g

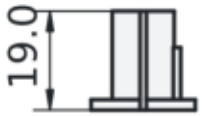
Filament: ~USD 0.05

Infill: 100%

None intended; inspect slicer preview

Photo-derived gear fit: print these two small pieces first and check against the actual gear and shaft.

Keyed sleeve OD11.9; flange OD20 x 2
D shaft clearance 6.3; flat at Y=2.65
Key 2.8 wide; outer X=7.2; height 10
Sleeve height 17 above 2 mm flange
Split 1.2; clamp cap compresses sleeve
Gear bore/key are photo-derived; test fit first
Print flange down; 100% infill; no supports



Front



Side

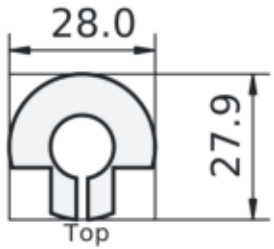
STL: gear adapter/gear_adapter_sleeve_r06.stl

Gear Adapter Clamp Cap R06

Print one • Native FreeCAD TechDraw: three orthographic views + isometric • Drawing shows local assembly orientation

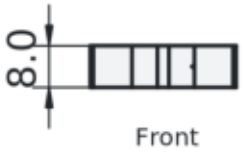
GEAR ADAPTER CLAMP CAP R06 | mm | r06 FIT CHECK

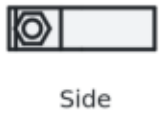
Mounting geometry inferred from references; verify with hardware.

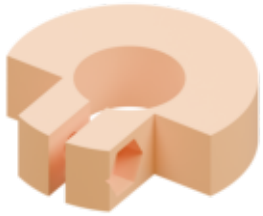




Cap OD28 / sleeve bore D12.2 / height 8
Split 2.2; M3 round bore D3.4
Captive nut pocket AF5.8 x 2.8 deep
Clamp bolt M3 x 16; standard M3 hex nut
Nut enters from outer right-hand face
Fit cap after sliding gear onto keyed sleeve
Round bores retain material around fasteners
100% infill; remove bore/nut-pocket supports







PRINT ORIENTATION

Flat annular face on bed; shaft opening vertical

PLA: ~4.0 g

Filament: ~USD 0.08

Infill: 100%

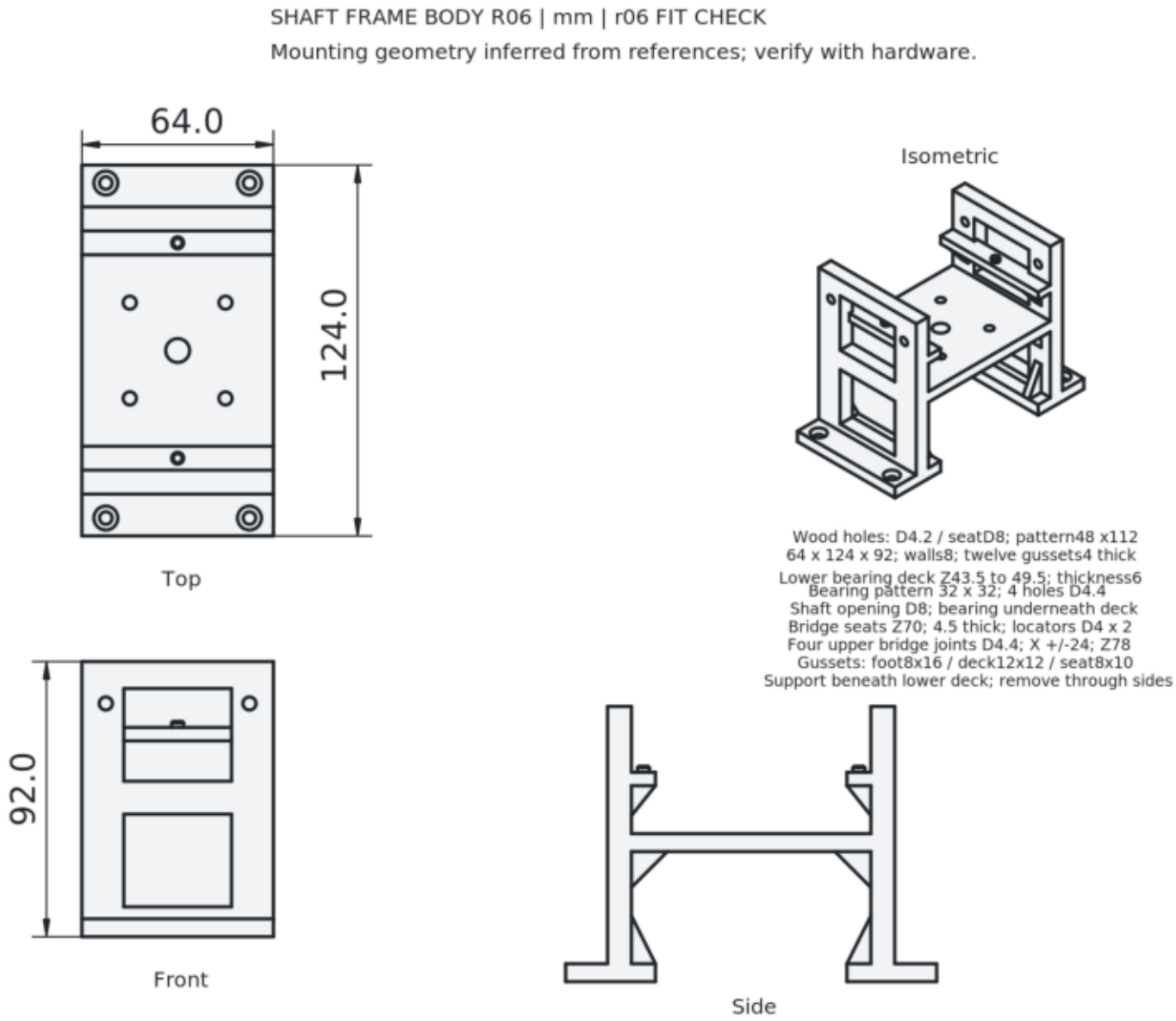
Support as needed in horizontal M3 passage and nut-pocket ceiling; remove fully before fastening

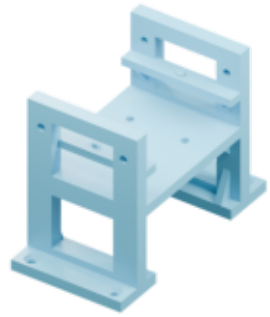
Photo-derived gear fit: print these two small pieces first and check against the actual gear and shaft.

STL: gear adapter/gear_adapter_clamp_cap_r06.stl

Shaft Frame Body R06

Print one • Native FreeCAD TechDraw: three orthographic views + isometric • Drawing shows local assembly orientation





PRINT ORIENTATION

Both integrated feet on bed; lower bearing deck horizontal

PLA: ~108.7 g

Filament: ~USD 2.17

Infill: 40%

Support beneath the lower bearing deck and overhanging seats; local support in horizontal M4 bores. Remove through the open X sides before fitting bearings.

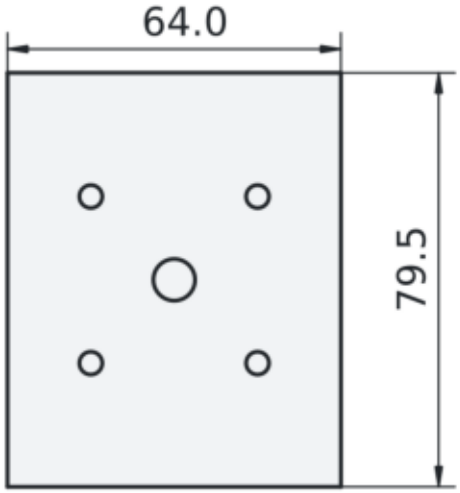
STL: shaft frame/shaft_frame_body_r06.stl

Shaft Upper Bearing Bridge R06

Print one • Native FreeCAD TechDraw: three orthographic views + isometric • Drawing shows local assembly orientation

SHAFT UPPER BEARING BRIDGE R06 | mm | r06 FIT CHECK

Mounting geometry inferred from references; verify with hardware.



Top

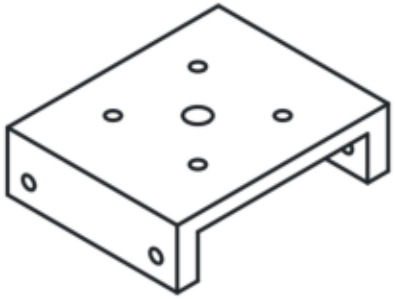


Front

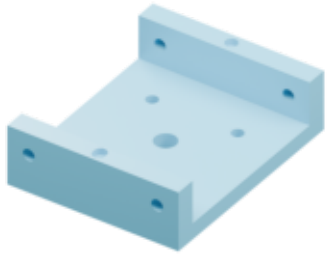


Side

Isometric



Bridge 64 x 79.5 x 22; deck thickness6
4 bearing holes D4.4 on 32 x 32 pattern
Shaft clearance D8; bearing above bridge
4 side joints D4.4; X +/-24; local Z8
End clearance 0.25 each side (assumed)
Front socket D4.5 x 2.25; rear slot4.5 x5.5
Locators align; shoulders carry bridge load
4 M4x25 + nuts; bearing uses 4 M4x16
Load both pulleys and belts before closing



PRINT ORIENTATION

Broad top deck on bed; edge flanges and locator recesses upward

PLA: ~42.1 g

Filament: ~USD 0.84

Infill: 40%

Local support as needed in four horizontal M4 joining bores; keep locator recesses and bearing faces clean.

STL: shaft frame/shaft_upper_bearing_bridge_r06.stl

Assembly: preserve alignment, then tension both belts

Motor 20T → gear 80T reduces speed; gear 80T → encoder 20T restores motor RPM at the encoder

1. Remove supports from the eight holder washer lands, horizontal round fastener bores and clamp nut pocket. Check the motor holes and encoder mounting group. Fit the motor with four M3 x 6 screws and encoder with three M3 x 8 screws, using seven thin washers. Retain case fasteners.

2. Before mounting the frame to plywood, fit the lower bearing under the integral lower deck and prefit the upper bearing to its removable bridge. Use eight M4 x 16 screws and eight plain nuts. Follow the illustrated bearing-sandwich page; leave bearing fasteners loose enough for alignment.

3. With the upper bridge removed, install the bare shaft, both 80T pulleys and both belts. Lower the bridge and its bearing along the bare shaft onto the locating pins and seating shoulders. Start four M4 x 25 screws from front/rear with four plain nuts accessed through the open sides; tighten evenly after alignment.

4. Fit the lower collar beneath its bearing and the upper collar above its bearing. Face the raised collar surfaces toward the bearing inner rings. Bring them into light contact and tighten without forcing axial preload. Check that the shaft rotates freely.

5. Insert captive slide nuts, then fix both slide bases and the integrated frame feet to plywood with twelve #6 x 1/2 inch flat-head wood screws. Fit the holders to their bases using eight M4 x 12 clamps and eight 1 mm washers. The lower bearing must already be installed because its screw access is from below.

6. Align the 20T motor pulley with the lower belt plane at 61 mm and the 20T encoder pulley with the upper plane at 77 mm. Nominal centers are 72.47 mm; each slide has 15 mm travel. Set only the tension needed for reliable tracking, and tighten the slide clamps evenly.

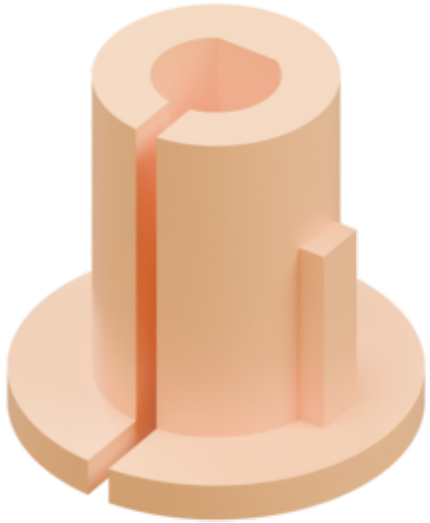
7. Assemble and fit-check the two-piece gear adapter as on the next page. Turn the mechanism by hand through several revolutions, checking shaft grip, pulley/fastener clearance and belt tracking. Evaluate vibration, stiffness and motor heat in a low-load trial before increasing load.

Keep the internal frame, bearing mounts and slides rigid. If vibration isolation is needed later, the preferred trial is TPU feet beneath the whole plywood board; any mating gear should share the same support. No TPU or soft bolt washers are included in this release.

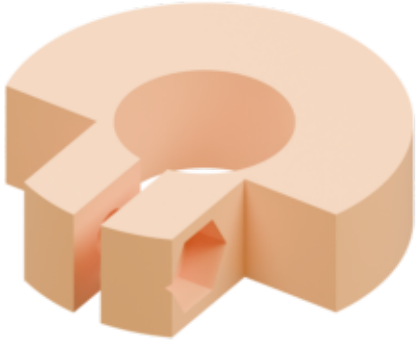
Fit the gear adapter before printing the large parts

Two pieces allow assembly around the existing gear • Gear bore and key dimensions are inferred from photographs

KEYED SLEEVE + SUPPORT FLANGE



REMOVABLE CLAMP CAP



1. Check the sleeve slides onto the actual 6 mm D shaft and that its 11.9 mm outside diameter and 2.8 mm key fit the gear. The reference bore is approximately 12 mm. Do not force the gear onto a tight print; adjust the source fit dimensions if necessary.

2. Slide the sleeve onto the shaft with its flange beneath the gear. Slide the gear over the keyed sleeve, then fit the separate cap over the exposed upper sleeve neck. The cap rests above the gear and can be positioned to match its actual thickness.

3. Insert the M3 nut from the right outer face into the hex pocket. Install the M3 x 16 screw from the left and tighten gradually until the cap compresses the split sleeve onto the shaft. Check grip against both twisting and axial movement by hand. Do not crush the plastic or close the split fully to compensate for a poor fit.

Both pieces use 100% infill. Shaft bore: 6.3 mm, flat at Y=2.65; splits: sleeve 1.2 mm, cap 2.2 mm. The cap has a round 3.4 mm M3 passage and full hex nut pocket; use and remove supports as needed. Clearances are design assumptions. Gear torque, axial load and long-term PLA clamp durability remain unverified.

Dimension sources and limits of this fit set

The encoder pattern previously fitted; new motor centers and the new adapter still need physical fit checks

Motor reference measurement — R0004472.JPG

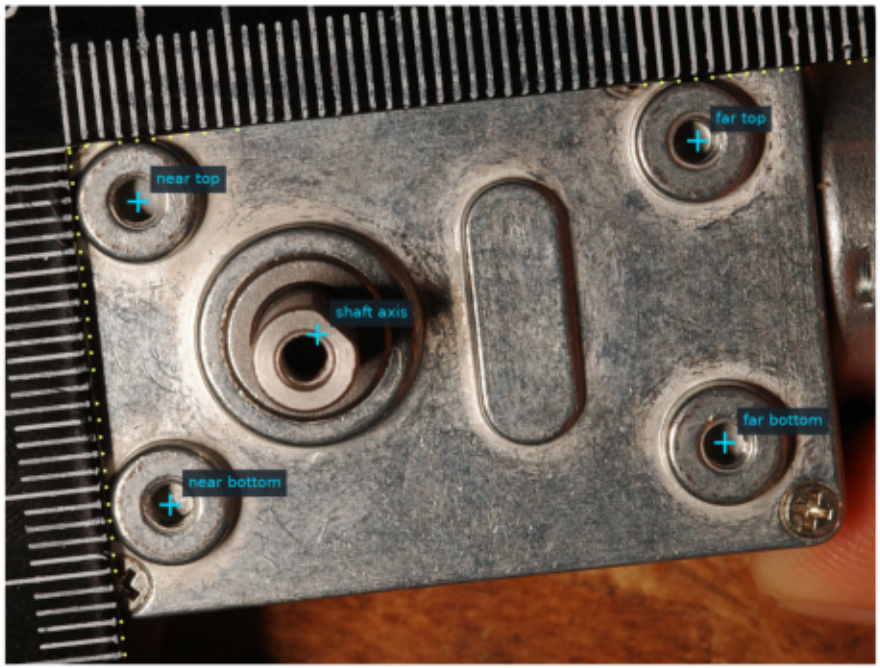


Photo estimates: pitch 33.06 × 17.86 mm; shaft offset 9.71 mm. Model: 33 × 18 mm / 9.7 mm.
58 ruler ticks, two-axis calibration. Photo estimates ≈±0.3 mm; FreeCAD drawings remain authoritative.

The leveled motor shaft-face photograph gives a 33 x 18 mm M3 hole pattern, with the shaft 9.7 mm from the nearer column. Nominal model holes are 3.8 mm round clearance holes; estimated photo uncertainty is about ±0.3 mm. This release preserves the corrected r05 mounting interface.

Encoder mounting slots retain the user-confirmed pattern. Gear bore/key, device body envelopes and pulley axial dimensions are still reference estimates. Valid solids, watertight meshes, hole/fastener clearances, collar access and slide travel have been checked; physical stiffness, vibration behavior and a working load limit have not been established.